



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Smart Renewable Energy Monitoring and Grid Optimization Platform

A CASE STUDY REPORT

Submitted in the partial fulfilment for the Course of

CSA1016 – SOFTWARE ENGINEERING

to the award of the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

Submitted by

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Smart Renewable Energy Monitoring and Grid Optimization Platform

1. Project Vision and Stakeholders

This platform is designed to continuously monitor renewable energy resources such as solar, wind, hydro, and battery storage while optimizing power distribution across the smart grid. The vision is to improve energy efficiency, reduce transmission losses, ensure reliable electricity supply, and support sustainable development. Stakeholders include grid operators, utility companies, renewable energy providers, maintenance engineers, administrators, consumers, government agencies, and researchers. The platform provides a centralized dashboard for monitoring performance, detecting faults, analyzing energy demand, forecasting renewable generation, and recommending optimized grid operations. Agile methodology ensures iterative development through sprint planning, daily stand-up meetings, backlog refinement, sprint review, and retrospective meetings. Continuous integration, automated testing, stakeholder feedback, risk management, and quality assurance improve software quality, maintainability, scalability, reliability, and customer satisfaction while supporting future enhancements.

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2. Epics and User Stories

Major epics include User Management, Renewable Energy Monitoring, Grid Optimization, Demand Forecasting, Alert Management, Reporting, and Predictive Maintenance. Example user stories include: As an administrator, I want secure login so that only authorized users access the system. As a grid operator, I want to monitor real-time generation so that I can balance supply and demand. As a maintenance engineer, I want instant fault alerts so that failures are resolved quickly. As a consumer, I want energy consumption reports so that I can reduce electricity usage. Agile methodology ensures iterative development through sprint planning, daily stand-up meetings, backlog refinement, sprint review, and retrospective meetings. Continuous integration, automated testing, stakeholder feedback, risk management, and quality assurance improve software quality, maintainability, scalability, reliability, and customer satisfaction while supporting future enhancements.

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3. Acceptance Criteria

The login module shall authenticate valid users within three seconds. The monitoring dashboard shall update sensor values in real time. Fault detection shall generate alerts whenever abnormal conditions occur. Optimization recommendations shall be displayed based on predicted demand and renewable energy availability. Reports shall be generated

accurately with export functionality. All modules shall satisfy functional and non-functional requirements including reliability, security, scalability, and usability. Agile methodology ensures iterative development through sprint planning, daily stand-up meetings, backlog refinement, sprint review, and retrospective meetings. Continuous integration, automated testing, stakeholder feedback, risk management, and quality assurance improve software quality, maintainability, scalability, reliability, and customer satisfaction while supporting future enhancements.

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4. Product Backlog Prioritization

High-priority features include authentication, dashboard, live monitoring, alerts, and energy optimization. Medium-priority features include AI forecasting, analytics, historical reports, and maintenance scheduling. Low-priority features include profile customization, feedback, themes, and multilingual support. Backlog refinement is performed during every sprint to ensure maximum customer value and continuous improvement. Agile methodology ensures iterative development through sprint planning, daily stand-up meetings, backlog refinement, sprint review, and retrospective meetings. Continuous integration, automated testing, stakeholder feedback, risk management, and quality assurance improve software quality, maintainability, scalability, reliability, and customer satisfaction while supporting future enhancements.

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5. Sprint Backlog

Sprint activities include requirement analysis, architecture design, UI implementation, database development, API integration, IoT sensor connectivity, dashboard creation, alert module implementation, testing, debugging, documentation, sprint review, retrospective, and deployment preparation. Daily stand-up meetings help monitor progress while sprint reviews validate completed features. Agile methodology ensures iterative development through sprint planning, daily stand-up meetings, backlog refinement, sprint review, and retrospective meetings. Continuous

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6. Story Point Estimation

Story points are estimated using the Fibonacci sequence. Login Module-3, Dashboard-5, Monitoring-8, Alerts-5, Reporting-5, AI Forecasting-13, Grid Optimization-13, Predictive Maintenance-8. Estimation considers implementation complexity, development effort, technical risks, dependencies, and testing requirements. Agile methodology ensures iterative development through sprint planning, daily stand-up meetings, backlog refinement, sprint review, and retrospective meetings. Continuous integration, automated testing, stakeholder feedback, risk management, and quality assurance improve software quality, maintainability, scalability, reliability, and customer satisfaction while supporting future enhancements.

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7. Sprint Goal and Expected Deliverables

The sprint goal is to deliver a functional prototype capable of secure authentication, renewable energy monitoring, intelligent grid optimization, alert management, and reporting. Expected deliverables include source code, database schema, test cases, user documentation, sprint review presentation, deployment guide, and demonstration of the completed prototype. Agile methodology ensures iterative development through sprint planning, daily stand-up meetings, backlog refinement, sprint review, and retrospective

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